

YP65AT BLE Module Specification



Contents

1 Product Overview.....	4
2 application area.....	4
3 Hardware Dimension Diagram.....	4
4 Pin Definition.....	4
5 electrical character.....	6
6 power dissipation.....	6
7 RF characteristics.....	7
8 Application Block Diagram	1
9 UART default setting.....	1
10 Power-On Sequence and Precautions.....	1
11 Default Service Description.....	9
11.1 Private service data transmission and reception.....	9
12 AT Protocol Summary.....	10
12.1 CMD gather.....	10
12.1.1 Configure or query the instruction set summary.....	10
12.1.2 Private service for sending custom data summaries.....	1
12.2 EVENT gather.....	12
12.2.1 Status Notification Event Summary.....	12
12.2.2 The private service receives a summary of custom data events.....	12
13 Detailed Description of AT CMD Configuration Commands.....	13
13.1 Test command AT\r\n.....	13
13.2 Check current basic parameters: AT+RX\r\n.....	13
13.3 Module Reset Command: AT+RESET\r\n.....	13
13.4 State Query Command: AT+STATE\r\n.....	13
13.5 Version Query Command: AT+VERSION\r\n.....	14
13.6 Bluetooth Name Query Command: AT+NAME=?\r\n.....	14
13.7 Bluetooth name configuration command: AT+NAME=YP65-AT\r\n.....	14
13.8 Query Bluetooth address: AT+ADDR=?\r\n.....	14
13.9 Set Bluetooth address: AT+ADDR=334455667788\r\n.....	14
13.10 Query current transmit power: AT+TXPWR=?\r\n.....	15
13.11 Set the transmission power: AT+TXPWR=0\r\n.....	15
13.12 Query serial port baud rate: AT+BAUD=?\r\n.....	15
13.13 Set the serial port baud rate: AT+BAUD=115200, N\r\n.....	15
13.14 Query broadcast data: AT+ADVDA=?\r\n.....	16
13.15 Set broadcast data: AT+ADVDA=020102\r\n.....	16
13.16 Query broadcast interval: AT+ADVINT=?\r\n.....	16
13.17 Set broadcast interval: AT+ADVINT=100\r\n.....	16
13.18 Query whether broadcasting is enabled: AT+ADVEN=?\r\n.....	17
13.19 Enable broadcast: AT+ADVEN=1\r\n.....	17
13.20 Disable broadcast: AT+ADVEN=0\r\n.....	17
13.21 Check whether the RTC function is enabled: AT+RTCEN=?\r\n.....	17

13.22	Enable RTC functionality: AT+RTCCEN=1\r\n.....	17
13.23	Disable RTC functionality: AT+RTCCEN=0\r\n.....	17
13.24	Set RTC timestamp: AT+RTCCDA=6955F140\r\n.....	18
13.25	Query RTC timestamp: AT+RTCCDA=?\r\n.....	18
13.26	Query sleep mode: AT+SLEEP=?\r\n.....	18
13.27	Set to full-speed operation mode: AT+SLEEP=0\r\n.....	18
13.28	Set BLE low-power mode: AT+SLEEP=1\r\n.....	18
13.29	Set Deep Sleep Mode: AT+SLEEP=2\r\n.....	19
14	AT CMD Data Transmission Command Detailed Specifications.....	19
14.1	In private service with UUID=0xfffl, sending custom data.....	19
14.2	Send custom data via private service (UUID=0xffl2).....	19
14.3	Send custom data via private service (UUID= 0xffl3).....	19
14.4	Send custom data via private service (UUID=0xffl4).....	19
14.5	In private service with UUID=0xffl5, sending custom data.....	20
15	Detailed Description of AT EVENT Status Notification Events.....	20
15.1	Bluetooth connection event notification: AT+CONNECT\r\n.....	20
15.2	Bluetooth disconnection event notification: AT+DISCONN\r\n.....	20
15.3	Module startup/Deep sleep wake-up event notification: AT+IAMREADY\r\n.....	20
15.4	Module enters deep sleep event notification: AT+POWERDWN\r\n.....	20
16	AT EVENT Data Transmission Event Detailed Description.....	1
16.1	Private service with UUID=0xfffl received custom data.....	1
16.2	Private service (UUID=0xffl2) receives custom data.....	1
16.3	Private service with UUID=0xffl3 received custom data.....	1
16.4	Private service with UUID=0xffl4 received custom data.....	1
16.5	Private service with UUID=0xffl5 received custom data.....	1

1 Product Overview

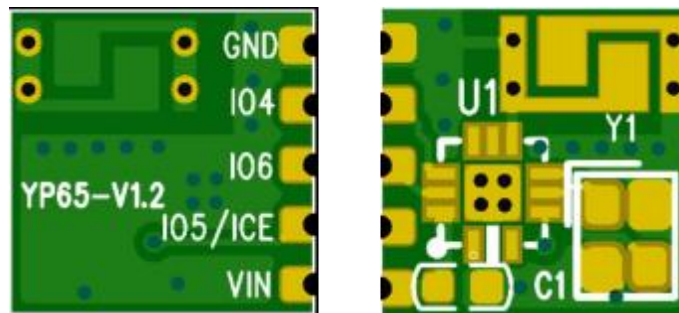
The YP65 is a ultra-compact Bluetooth BLE passthrough module developed by Yiprinciple Technology based on the Yizhao Micro Bluetooth chip. The YP65 is a BLE module compliant with the Bluetooth 5.4 standard; it is built upon a single-core Bluetooth chip from the supplier YICHIP and adheres to the Bluetooth 5.4 specification.

The wireless operating frequency is 2.4 GHz, the modulation scheme is GFSK, and the air interface data rate is 1 Mbps. The maximum transmit power of the module is 5 dBm. The HID function can be enabled or disabled; when enabled, it supports pairing and connection with BLE-enabled iOS and Android devices. It complies with BQB, CE, FCC, KC, and SRRC certifications. Users can modify parameters such as the serial port baud rate and device name according to their needs. The module supports full-speed operation mode, intermittent low-power mode, and deep sleep/long sleep mode.

2 Application Area

YP65 supports the latest Bluetooth standard BLE 5.4, enabling pairing and connectivity with iOS and Android devices that support BLE.

- New National Standards for chargers and power banks;
- Mobile phone accessories;
- Computer peripheral devices;
- Wireless data transmission for medical devices;
- Wireless data transmission for in-vehicle instruments.



3 Hardware Dimension Diagram

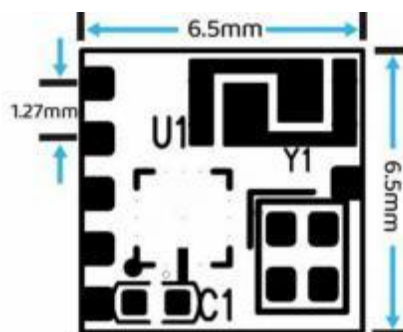


Figure 1: YP65 Hardware Dimensions Diagram

4 Pin Definition

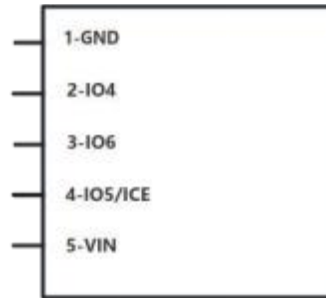


Figure 2: YP65 Pin Definitions

Pin Number	Name	Type	Fuction
1	GND	Power/Ground	Systematically, reference to zero potential.
2	I04	DIO	BLE-TX – Module serial port transmitter.
3	I06	DIO	BLE-RX: Module serial port receiver.
4	I05/ICE	DIO	Wakeup IO wake-up signal (low-to-high transition for wake-up/high-to-low transition for sleep); AT/DATA selection (valid only in AT command mode); Bluetooth ICE debugging interface.
5	VIN	PWR	Main power input: 1.8–4.3 V.

Note: GPIO5 has a default software pull-up configuration.

1. When the module is operating at full speed (sleep mode = 0), IO5 merely serves as the pin for switching the Transparent Command Configuration Command:

- 1.1. When the IO5 pin is pulled low, the serial port data transmitted represents Bluetooth connection status data, effectively forwarding Bluetooth data.
- 1.2. For high-speed serial transmission, IO5 represents serial port data as AT configuration commands.

2. When the module is in the intermittent sleep mode (sleep mode = 1), IO5 must first serve as the wake-up pin, and then serve as the switching pin for both the transparent command and the configuration command:

- 2.1. The MCU sends an AT configuration command: first, pull the wake-up pin (GPIO5) low for at least 10 ms to activate the wake-up function; then, release GPIO5 to set it to a high level, indicating that the system is currently in the AT command mode.
- 2.2. MCU sends Bluetooth passthrough data: First, pull the wake-up pin (GPIO5) low for at least 10 ms to activate the wake-up function; subsequently, maintain the GPIO5 pin in a low state to indicate that the system is currently in the AT data transmission mode.

Note: Both the 2.1 and 2.2 processes will reset the sleep mode to 0 upon waking up; therefore, if the module needs to continue operating in intermittent sleep mode, it is necessary to release GPIO5 to set it to a high logic level and then resend the AT+SLEEP=1\r\n command.

Note: The 'sleep mode=1' configuration represents a reserved sleep mode that supports IO wake-up and timed wake-up; upon waking, the configuration information will not be lost.

3. When the module is in deep sleep mode (sleep mode=2), GPIO5 must first serve as a wake-up pin, and then as a switching pin for both the transmit command and the configuration command:

- 3.1. Send the AT configuration command: Continuously pull the wake-up pin (GPIO5) low to activate the wake-up function; wait until the serial port receives the response 'AT+IAMREADY\r\n', indicating that the module has been successfully woken up; then, pull GPIO5 high and send the AT configuration command.

3.2. Sending Bluetooth transparent data: Since the connection is invalid and the device wakes up from deep sleep, the Bluetooth module will revert to its initial state (no broadcasting, no connection) and then return to the full-speed operation state described in the previous section.

Note: The 'sleep mode=2' configuration represents a deep sleep mode that only supports IO wake-up; upon waking, all configuration information will be lost.

5 Electrical Character

Rating	Min	Typ	Max	Unit
VIO		3.3	3.6	V
VIN/HVIN	1.8	3.3	5.5	V
Ambient temperature	-15	/	+85	°C

6 Power Dissipation

W/O DC-DC	Parameter	Average	Unit
Sleep	/	1.5	uA
Sniff	110ms interval	135	uA
Discoverable	ADV interval: 100ms	356	uA
	ADV interval: 1000ms	40	uA

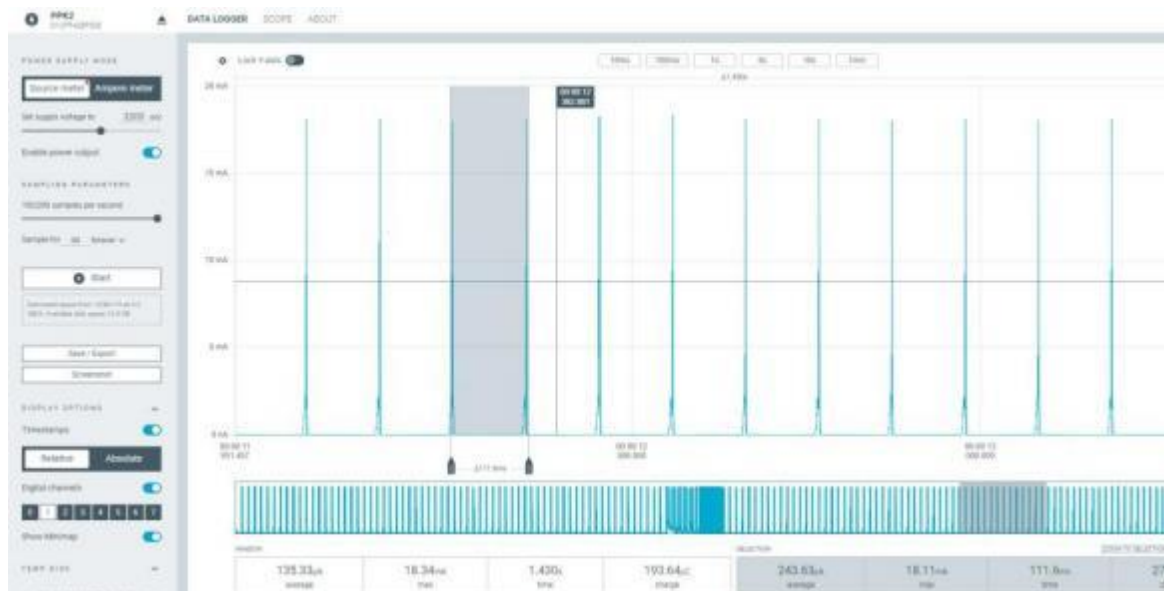




Figure 3: Power Consumption

7 RF Characteristics

Rating	Value	Unit
BLE transmitting power	5	dBm
BLE sensitivity: 1 Mbps	-97	dBm

8 Application Block Diagram

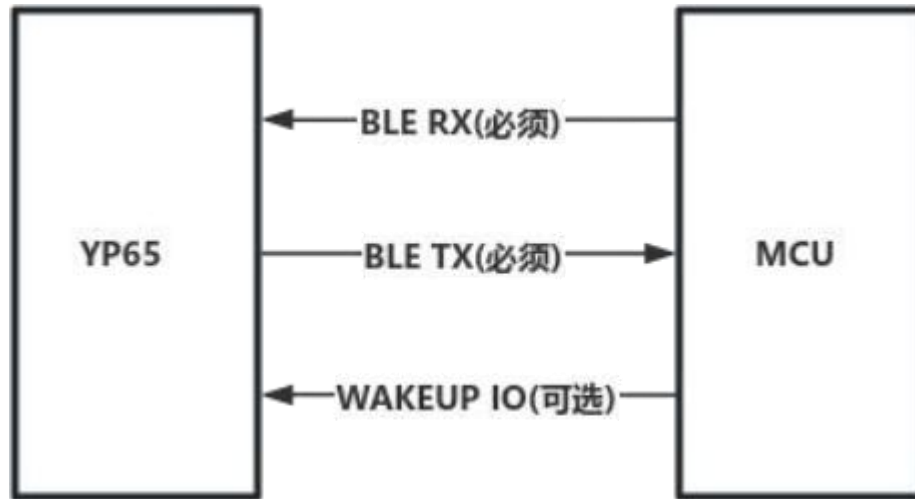


Figure 4: Module Application Block Diagram

9 UART Default Setting

Baud rate:	115200
data bit:	8
stop bit:	1
check bit:	not have
Flow control:	None

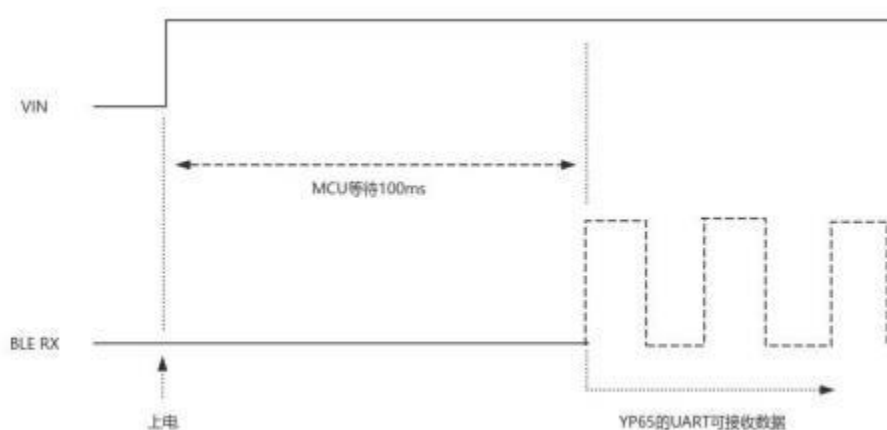
10 Power-On Sequence and Precautions

The MCU waits for the YP65 to complete the boot process, then downloads the patch via the serial port.

Note: The BOOT time for YP65 is 50ms+, so a relatively safe waiting period is approximately 100ms.

The MCU's operating timing is shown in the table below:

	YP65 Power-On		BOOT completed – Download PATCH
Power-On Operation Sequence	YP65 – VIN Power-On	Wait 100ms	UART Download Patch



11 Default Service Description

11.1 Private Service Data Transmission and Reception

Service UUID: 0xFFFF0

UUID	Attribute	Service Handle ID
0xFFFF1	Notify+Write No Response	0x001b
0xFFFF2	Notify+Write No Response	0x001e
0xFFFF3	Notify+Write No Response	0x0021
0xFFFF4	Notify+Write No Response	0x0024
0xFFFF5	Notify+Write No Response	0x0027

Among these, "Notify" is used for data uplink, while "Write No Response" is used for data downlink.

12 AT Protocol Summary

12.1 CMD Gather

12.1.1 Configure or Query the Instruction Set Summary

The MCU sends AT-format serial data to the BLE chip, which is commonly used for configuring Bluetooth parameters or querying the Bluetooth status.

Order number	ATcmd instruct	Cmd res	Act on	Windows default
1	AT\r\n	OK\r\n	Test instruction	Null
2	AT+RX\r\n	+NAME:YP65-AT\r\n +ADDR:2A 07 98 20 28 66\r\n +TXPWR:0dbm\r\n +STATE:00 00\r\n +BAUD:115200,NONE\r\n	View current basic parameters	Null
3	AT+RESET\r\n	OK\r\n	Module Restart Command	Null
4	AT+STATE\r\n	+ STATE:00 02\r\n + STATE:00 08\r\n	Check module status (Bit 9); Bluetooth connection status (Bit 11); Bluetooth broadcast status	00 00
5	AT+VERSION\r\n	+VERSION:V0.01\r\n	Check software version	V0.01
7	AT+NAME=?\r\n	+NAME:YP65-AT\r\n	View Bluetooth Name	YP65-AT
	AT+NAME= 111\r\n	OK\r\n	Change the Bluetooth name to "111", with a maximum length of 29 bytes.	Null
8	AT+ADDR=?\r\n	+ADDR:2A 07 98 20 28 66\r\n	View Bluetooth Address	Null
	AT+ADDR= 334455667788\r\n	OK\r\n	Change the Bluetooth MAC address to 33:44:55:66:77:88	Null
9	AT+TXPWR=?\r\n	+TXPWR:5dbm\r\n	View RF power	0dbm
	AT+TXPWR=0\r\n	OK\r\n	Set the RF power to 5 dB	Null
	AT+TXPWR=1\r\n	OK\r\n	Modify RF power command: 0 dB	Null
	AT+TXPWR=2\r\n	OK\r\n	Modify RF power command: -5 dB	Null
	AT+TXPWR=3\r\n	OK\r\n	Modify RF power command – 10 dB	Null
10	AT+BAUD=?\r\n	+BAUD:9600,NONE\r\n	Check serial port baud rate	9600,NO NE
	AT+BAUD= 115200,E\r\n	OK\r\n	Set the serial port baud rate to 115200; set the parity mode to EVEN.	Null
	AT+BAUD= 19200,O\r\n	OK\r\n	Set the serial port baud rate to 19200; set the parity mode to ODD.	Null
	AT+BAUD=9600,N\r\n	OK\r\n	Set the serial port baud rate to 9600 – No parity (Default)	Null

Order number	ATcmd instruct	Cmd res	Act on	Windows default
11	AT+ADVDA=?\r\n	+ADVDA:02 01 06\r\n	View broadcast data	
	AT+ADVDA=02010203FF0101\r\n	OK\r\n	Modify the broadcast data to 02 01 02 03 FF 01 01 (HEX)	Null
13	AT+ADVINT=?\r\n	+ADVINT:160\r\n	View module broadcast interval	160
	AT+ADVINT= 160\r\n	OK\r\n	Set the module broadcast interval to $160 \times 0.625 \text{ ms} = 100 \text{ ms}$; valid range: $32-16,000 \times 0.625 \text{ ms}$; default value: $320 (320 \times 0.625 \text{ ms} = 200 \text{ ms})$.	Null
15	AT+ADVEN=?\r\n	+ADVEN:1\r\n	Check whether the module is broadcasting.	0
	AT+ADVEN=0\r\n	OK\r\n	Disable module broadcasting	Null
	AT+ADVEN=1\r\n	OK\r\n	Enable Module Broadcast	Null
16	AT+RTCEN=?\r\n	+RTCEN:0\r\n	Check whether the module is enabled for RTC counting.	0
	AT+RTCEN=0\r\n	OK\r\n	Disable RTC timer module	Null
	AT+RTCEN=1\r\n	OK\r\n	Enable RTC timing module	Null
17	AT+RTCDA=?\r\n	+RTCDA: 6955F140\r\n	View RTC Timing Module	0x6955F140
	AT+ RTCDA = 6955F140\r\n	OK\r\n	0x6955F140 = 1,767,240,000, means 2026-01-01 12:00:00	Null
18	AT+SLEEP=0\r\n	OK\r\n	Turn off the sleep mode of the module	0
	AT+SLEEP=1\r\n	OK\r\n	Module enters sleep mode	Null
	AT+SLEEP=2\r\n	OK\r\n +AT+POWERDOWN\r\n	Module enters shutdown mode	Null

12.1.2 Private Service for Sending Custom Data Summaries

Note: Data can be sent directly in HEX format: a 2-byte handle identifier followed by the data. The table below provides an example of how to send data using 5 UUIDs that support `notiFy`.

Order number	UUID	HANDLE	MCU→YP65 (uart)	Explain
1	0xfff1	0x001b	2a 00 01 01 02 03	Send data to the mobile phone via handle 0x002a (UUID = 0xFFF1).
2	0xfff2	0x001e	2d 00 01 01 02 03	Send data to the mobile phone at handle 0x002d (UUID = 0xFFF2).
3	0xfff3	0x0021	30 00 01 01 02 03	Send data to the mobile phone via handle 0x0030 (UUID = 0xFFF3).
4	0xfff4	0x0024	33 00 01 01 02 03	Send data to the mobile phone at handle 0x0033 (UUID = 0xFFF4).
5	0xfff5	0x0027	36 00 01 01 02 03	Send data to the mobile phone at handle 0x0036 (UUID = 0xFFF5).

pay attention to :

1. Local MTU default setting: 185; upon connection, the Bluetooth module actively exchanges the MTU with the mobile phone.
2. Bluetooth connection parameter request: Minimum connection interval = 8×1.25 ms; Maximum connection interval = $32 \times 1.25 = 40$ ms; Latency count = 5; Connection timeout duration = 500×10 ms; Upon establishing a connection, the YP65 Bluetooth module will proactively update its Bluetooth connection parameters.
3. Two consecutive AT commands must be separated by at least 20 ms to allow the Bluetooth module to separate the packets; specifically, there must be a minimum delay of 20 ms between the last byte of the first packet and the first byte of the second packet.

12.2 EVENT Gather

12.2.1 Status Notification Event Summary

Order number	AT-Event Command	Explain
1	AT+CONNECT\r\n	Bluetooth connection event
2	AT+DISCONN\r\n	Bluetooth Disconnection Event
3	AT+POWERDWN\r\n	The YP65 chip enters deep sleep mode.
4	AT+IAMREADY\r\n	YP65 chip cold boot or wake-up from deep sleep

12.2.2 The Private Service Receives a Summary of Custom Data Events.

Note: HEX format: 2-byte handle + data.

The table below shows an example of sending data to 5 UUIDs that support "write no response":

Order number	Event (HEX format)	Instance	Explain
1	1b 00 xx xx xx	1b 00 01 02 03	Received data 01 02 03 at Handle value 0x001B (UUID = 0xfff1).
2	1e 00 xx xx xx	1e 00 11 22 33 44	Data 11 22 33 44 received at Handle ID 0x001e (UUID = 0xfff2)
3	21 00 xx xx xx	21 00 01 02 03	Received data 01 02 03 at Handle value 0x0021 (UUID = 0xfff3).
4	24 00 xx xx xx	24 00 01 02 03	Received data 01 02 03 at Handle value 0x0024 (UUID = 0xfff4).
5	27 00 xx xx xx	27 00 01 02 03	At handle 0x0027 (UUID = 0xFFF5), data 01 02 03 was received.

Note:The MCU receives data via the serial port; to facilitate packet unpacking by the MCU, it is recommended that the data packet format sent by the APP includes the packet length information.

13 Detailed Description of AT CMD Configuration Commands

13.1 Test Command AT\r\n

Instruction Description
Format: AT\r\n Response: OK\r\n

13.2 Check Current Basic Parameters: AT+RX\r\n

Instruction Manual
Format: AT+RX\r\n Respond : +NAME:YP65-AT\r\n +ADDR:2A 07 98 20 28 66\r\n +TXPWR:0dbm\r\n +STATE:00 00\r\n +BAUD:115200,NONE\r\n

13.3 Module Reset Command: AT+RESET\r\n

Instruction Manual
Format: AT+RESET\r\n Response: OK\r\n

13.4 State Query Command: AT+STATE\r

Instruction Manual
Format: AT+STATE\r\n Response: +STATE: 00 00\r\n

Note: When Bit9 is set to 1, it indicates that Bluetooth is connected; when Bit9 is set to 0, it indicates that Bluetooth is not connected. When Bit11 is set to 1, it indicates that broadcasting is enabled; when Bit11 is set to 0, it indicates that broadcasting is disabled.

Default value: 00 00

13.5 Version Query Command: AT+VERSION\r\n

Instruction Manual
Format: AT+VERSION\r\n Response: +VERSION: V0.01\r\n

13.6 Bluetooth Name Query Command: AT+NAME=?\r\n

Instruction Manual
Format: AT+NAME=?\r\n Response: +NAME: YP65-AT\r\n

Default value: YP65-AT

13.7 Bluetooth Name Configuration Command: AT+NAME=YP65-AT\r\n

Instruction Manual
Format: AT+NAME=YP65-AT\r\n Response: OK\r\n

Note: The Bluetooth device name can be up to 29 bytes long; this command updates the name only in the Scan Response and Bluetooth Service List. To add a name to the Advertising Data, use the AT+ADVDA=x command.....`x\r\n` command.

13.8 Query Bluetooth MAC Address: AT+ADDR=?\r\n

Instruction Manual
Format: AT+ADDR=?\r\n Response: +ADDR: 2A 07 98 20 28 66\r\n

Note: The default MAC can be generated by the burn-in tool using a random code generator.

13.9 Set Bluetooth Address: AT+ADDR=334455667788\r\n

Instruction Manual
Format: AT+ADDR=334455667788\r\n Response: OK\r\n

Note: For hexadecimal string representation, the module only recognizes characters A–F; it does not recognize characters a–f.

13.10 Query Current Transmission Power: AT+TXPWR=?\r\n

Instruction Manual

Format: AT+TXPWR=?\r\n

Response: +TXPWR: 0 dBm\r\n

Default value: 0 dBm.

13.11 Set Transmission Power: AT+TXPWR=0\r\n

Instruction Manual

Format: AT+TXPWR=0\r\n

Response: OK\r\n

AT+TXPWR=0\r\n indicates 5 dB.

AT+TXPWR= 1\r\n indicates 0 dB.

AT+TXPWR=2\r\n indicates -5 dB.

AT+TXPWR=3\r\n indicates -10 dB.

13.12 Query Serial Port Baud Rate: AT+BAUD=?\r\n

Instruction Manual

Format: AT+BAUD=?\r\n

Response: +BAUD: 115200, NONE\r\n

Default value: 115200, NONE

13.13 Set the Serial Port Baud Rate: AT+BAUD=115200, N\r\n

Instruction Manual

Format: AT+BAUD=115200, NONE\r\n

Response: OK\r\n

Serial port baud rate support: 9600–921600

Serial port parity check support: N, O, E

Other examples:

1. Set baud rate to 9600 and even parity:

form: AT+BAUD=9600,E\r\n

respond: OK\r\n

2. Set Baud Rate to 115,200; Odd Parity:

form: AT+BAUD=115200,O\r\n

respond: OK\r\n

3. Set Baud Rate to 921,600 (No Parity):

form: AT+BAUD=921600,N\r\n

respond: OK\r\n

13.14 Query Broadcast Data: AT+ADVDA=?\r\n

Instruction Manual

Format: AT+ADVDA=?\r\n

Response: +ADVDA: 02 01 06 03 03 12 18 03 19 C1 03

Default value: 02 01 06 03 03 12 18 03 19 C1 03

13.15 Set Broadcast Data: AT+ADVDA=020102\r\n

Instruction Manual

Format: AT+ADVDA=020102\r\n

Response: OK\r\n

Note: For hexadecimal string representation, the module only recognizes characters A–F; it does not recognize characters a–f.

Note: The broadcast length cannot exceed 31 bytes.

13.16 Query Broadcast Interval: AT+ADVINT=?\r\n

Instruction Manual

Format: AT+ADVINT=?\r\n

Response: +ADVINT= 160\r\n

Note: Unit * 0.625 ms; Example: 160 * 0.625 ms = 100 ms

Range: [32,16000)

Default value: 160

13.17 Set the Broadcast Interval: AT+ADVINT=100\r\n

Instruction Manual

Format: AT+ADVINT=100\r\n

Response: OK\r\n

13.18 Query Whether Broadcasting Is Enabled: AT+ADVEN=?\r\n

Instruction Manual

Format: AT+ADVEN=?\r\n

Response: +ADVEN=0\r\n

Note: 0 – Off, 1 – On; the broadcast function is disabled by default.

13.19 Enable Broadcast: AT+ADVEN=1\r\n

Instruction Manual

Format: AT+ADVEN= 1\r\n

Response: OK\r\n

13.20 Turn off Broadcasting: AT+ADVEN=0\r\n

Instruction Manual

Format: AT+ADVEN=0\r\n

Response: OK\r\n

13.21 Query the RTC Function Enable Status: AT+RTCEN=?\r\n

Instruction Manual

Format: AT+RTCEN=?\r\n

Response: +RTCEN:0\r\n

Note: 0 – Off; 1 – On; RTC functionality is disabled by default.

13.22 Enable RTC Functionality: AT+RTCEN=1\r\n

Instruction Manual

Format: AT+RTCEN= 1\r\n

Response: OK\r\n

13.23 Disable RTC Functionality: AT+RTCEN=0\r\n

Instruction Manual

Format: AT+RTCEN=0\r\n

Response: OK\r\n

13.24 Set RTC Timestamp: AT+RTCDA=6955F140\r \n

Instruction Manual

Format: AT+RTCDA=6955F140\r \n
Response: OK\r \n

Default initial value: 0x6955f140 (HEX) = 1767240000 (DEC) = 2026-01-01 12:00:00 (UTC)

13.25 Query RTC Timestamp: AT+RTCDA=?\r \n

Instruction Manual

Format: AT+RTCDA=?\r \n
Response: +RTCDA=6955F140\r \n

13.26 Query Sleep Mode: AT+SLEEP=?\r \n

Instruction Manual

Format: AT+SLEEP=?\r \n
Response: +SLEEP=0\r \n

Note: In the default sleep mode, the device operates at full speed without entering sleep mode.

13.27 Set to Full-Speed Operation Mode: AT+SLEEP=0\r \n

Instruction Manual

Format: AT+SLEEP=0\r \n
Response: OK\r \n

13.28 Set BLE Low-Power Mode: AT+SLEEP=1\r \n

Instruction Manual

Format: AT+SLEEP= 1\r \n
Response: OK\r \n

Note: Supports IO wake-up and timed wake-up; configuration information is not lost upon wake-up.

13.29 Set Deep Sleep Mode: AT+SLEEP=2\r\n

Instruction Manual

Format: AT+SLEEP=2\r\n
Response: OK\r\n

Note: Only IO wake-up is supported; all configuration information will be lost after waking up.

14 AT CMD Data Transmission Command Detailed Specifications

HEX format description: 2-byte handle value + specific data

Note: When sending data, keep GPIO5 at a low level to indicate that the system is currently in data transmission mode.

14.1 In Private Service With UUID=0xfff1, Sending Custom Data

Instruction Description (HEX)

Format: 1B 00 01 01 02 03
Respond : Null

14.2 Send Custom Data Via Private Service (UUID=0xfff2)

Instruction Description (HEX)

Format: 1e 00 01 01 02 03
Respond : Null

14.3 Send Custom Data Via Private Service (UUID = 0xfff3)

Instruction Description (HEX)

Format: 21 00 01 01 02 03
Respond : Null

14.4 Send Custom Data Via Private Service (UUID=0xfff4)

Instruction Description (HEX)

Format: 24 00 01 01 02 03
Respond : Null

14.5 In Private Service With UUID=0xfff5, Sending Custom Data

Instruction Description (HEX)
Format: 27 00 01 01 02 03 Respond : Null

15 Detailed Description of AT EVENT Status Notification Events

15.1 Bluetooth Connection Event Notification: AT+CONNECT\r\n

Instruction Manual
Format: AT+CONNECT\r\n Respond : Null

15.2 Bluetooth Disconnection Event Notification: AT+DISCONN\r\n

Instruction Manual
Format: AT+DISCONN\r\n Respond : Null

15.3 Module Startup/Deep Sleep Wake-up Event Notification: AT+IAMREADY\r\n

Instruction Manual
Format: AT+IAMREADY\r\n Respond : Null

15.4 Module Enters Deep Sleep Event Notification: AT+POWERDWN\r\n

Instruction Manual
Format: AT+POWERDWN\r\n Respond : Null

16 AT EVENT Data Transmission Event Detailed Description

Format Specification (HEX): 2-byte handle value + specific data

Note: The custom data transmitted by the APP must include parcel information, enabling the MCU to decompress the package upon receipt.

16.1 Private Service With UUID=0xffff1 Received Custom Data

Instruction Description (HEX)

Format: 1b 00 01 02 03

Respond : Null

Note: The handle 0x001b, corresponding to the private service UUID=0xffff1, received custom data: 01,02,03.

16.2 Private Service (UUID=0xffff2) Receives Custom Data.

Instruction Description (HEX)

Format: 1e00 01 02 03

Respond : Null

Note: The handle value 0x001e, corresponding to the private service UUID=0xffff2, receives custom data 01 02 03.

16.3 Private Service With UUID=0xffff3 Received Custom Data

Instruction Description (HEX)

Format: 21 00 01 02 03

Respond : Null

Note: The handle 0x0021, corresponding to the private service with UUID=0xffff3, received custom data: 01,02,03.

16.4 Private Service With UUID=0xffff4 Received Custom Data

Instruction Description (HEX)

Format: 24 00 01 02 03 04 05 06

Respond : Null

Note: The handle value 0x0024, corresponding to the private service UUID=0xffff4, receives custom data: 01 02 03 04 05 06.

16.5 Private Service With UUID=0xffff5 Received Custom Data

Instruction Description (HEX)

Format: 27 00 11 22 33 44 55

Respond : Null

Note: The handle value 0x0027, corresponding to the private service UUID 0xff5, receives custom data: 11 22 33 44 55.